

Lecture Objectives and Pre-Requisites

Key Takeaways

The key takeaways from this lecture are:

- Limitations of simple trading rules
- Introduction to autoregressive (AR) models for addressing the above problem
- Importance of stationarity
- ARIMA as an institutional approach
- The usage of dynamic modeling with rolling forecasts
- The usage of ACF and PACF for model order selection
- The significance of modeling returns, and not prices

Prerequisites

To follow along and implement the concepts covered in this lecture, ensure that you have the following installed in your environment:

- **numpy, pandas:** numerical/data operations
- **matplotlib, seaborn:** data visualization
- **yfinance:** financial data
- **datetime, time:** handling dates and timing execution
- **statsmodels:** ARIMA diagnostics, ADF test, and ACF/PACF plots

Please refer to the documentation of each library if you run into any issues during installation. In case you have trouble installing one or more of them, it'll be a minor annoyance for you during the lecture (since you won't be able to run some of the code). However, you should still be able to follow most of the material covered in the lecture.

Recommended Time for the Session & Related Coursework - 9 hours

Additional Reads (Blogs)

- **ARIMA Models:**

- <https://blog.quantinsti.com/autocorrelation-autocovariance/>

- <https://blog.quantinsti.com/autoregression/>

- <https://blog.quantinsti.com/autoregressive-moving-average-arma-model/>

- <https://blog.quantinsti.com/autoregressive-moving-average-arma-model-python/>

- <https://blog.quantinsti.com/autoregressive-moving-average-arma-model-r/>

- <https://blog.quantinsti.com/forecasting-stock-returns-using-arma-model/>

- **More Complex Models:**

- <https://blog.quantinsti.com/arfima-model/>

- <https://blog.quantinsti.com/artfima-model/>

- <https://blog.quantinsti.com/vector-autoregression/>

- <https://blog.quantinsti.com/tvp-var-stochastic-volatility/>